

Student Number _____ Name _____ Signature _____

This information will not be visible to the marker.



THE UNIVERSITY
of EDINBURGH

Financial Mathematics
MATH10003

Friday 10 May 2024
13:00–15:00

Number of questions: 4
Total number of marks: 100

PART A Please complete clearly

Exam Number
as shown on your university card

--

IMPORTANT PLEASE READ CAREFULLY

Before the examination

1. Put your university card face up on the desk.
2. **Complete PART A and PART B above.** By completing PART B you are accepting the University Regulations on student conduct in an examination (see back cover).
3. Complete the attendance slip and leave it on the desk.
4. This is a closed-book examination. No notes, printed matter or books are allowed.
5. A scientific calculator is permitted in this examination. It must not be a graphic calculator. It must not be able to communicate with any other device.

During the examination

1. Write clearly, in ink, in the space provided after each question. If you need more space then please use the extra pages at the end of the examination script or ask an invigilator for additional paper.
2. **Attempt all questions.**
3. If you have rough work to do, simply include it within your overall answer – put brackets at the start and end of it if you want to highlight that it is rough work.

At the end of the examination

1. This examination script must not be removed from the examination venue.
2. There are extra pages for working at the end of this examination script. If used, you should clearly label your working with the question to which it relates.
3. Additional paper and graph paper, if used, should be attached to the back of this examination script. Write your examination number on the top of each additional sheet.

[Do not write on this page]

1. Consider the one-period binomial model where a non-dividend-paying stock S has the spot price $S_0 = \text{€}18$ and in 1 year time it can either be $\text{€}27$ with probability 0.4 or $\text{€}9$ with probability 0.6. The risk-free discretely compounded interest rate per annum for one year is 5%.

- (a) i. Verify that the market is free of arbitrage. [3 marks]
ii. Recall that the risk-neutral measure is the probability measure $\mathbb{Q}$, which is determined by probability weights q_u and q_d , such that

$$S_0 = \frac{1}{1+r} \mathbb{E}^{\mathbb{Q}}[S_1]. \quad (1)$$

Using (1) and that $\mathbb{Q}$ is a probability measure, derive the formulae for q_u and q_d . Then find q_u and q_d for this market model. [6 marks]

If you have used additional space for working then please tick here: ☐

- (b) Consider a contingent claim Φ that has maturity $T = 1$ year and is written on the stock S . Furthermore, we know the payoff of Φ at maturity is given by

$$\begin{cases} S_T, & \text{if } S_T \leq 10, \\ 20 - S_T, & \text{if } 10 < S_T \leq 20, \\ S_T - 20, & \text{if } S_T > 20. \end{cases}$$

Meanwhile, suppose that we can buy and sell the following European options:

- European Call with strike $K_1 = \text{€}20$ and maturity $T = 1$ year.
 - European Put with strike $K_1 = \text{€}20$ and maturity $T = 1$ year.
 - European Put with strike $K_2 = \text{€}10$ and maturity $T = 1$ year.
- i. Draw a diagram of how the payoff of Φ change according to S_T , the stock price on the maturity date. [4 marks]

If you have used additional space for working then please tick here: ☐

- ii. Find a portfolio consisting of three European options listed above that replicates the claim Φ , and justify your answer. You should specify the position and the number of units of each option. (Hint: suppose there are x units $\text{Call}(T, K_1)$, y units $\text{Put}(T, K_1)$ and z units $\text{Put}(T, K_2)$ in this replicating portfolio, what is the payoff?) [9 marks]

If you have used additional space for working then please tick here: ☐

- iii. Using the replicating portfolio you found in the previous part and the Law of One Price, find the arbitrage-free price of the contingent claim Φ . Please round your answer to 3 decimal places. [8 marks]

[Total for this question: 30 marks]

If you have used additional space for working then please tick here: ☐

2. The price of the stock S today is £100, and it is expected to increase by 20% or decrease by 10% every 3 months over the next half year. The continuously compounded risk-free interest rate per annum is 4%.

- (a) i. Show that the market is free of arbitrage. [3 marks]
ii. Recall that under the continuously compounded interest rate, the risk-neutral measure $\mathbb{Q}$ consists of weights q_u and q_d such that

$$S_0 = e^{-r\Delta t} \mathbb{E}^Q[S_1].$$

Find the risk-neutral probabilities q_u and q_d for this two-period market. [4 marks]

If you have used additional space for working then please tick here: ☐

(b) Find the arbitrage-free price of a 6-month European call option Φ with strike price $K = \text{€}100$.

If you have used additional space for working then please tick here: ☐

- (c) Find the arbitrage-free price of a 6-month European-type put option Ψ with strike price $K = \text{€}100$, whose payoff function is

$$g(S_T) = ((K - S_T)^+)^2.$$

[7 marks]

If you have used additional space for working then please tick here: ☐

- (d) Find the arbitrage-free price of a 6-month American put option X with strike price $K = €110$.
Is there any early exercise opportunity?

[10 marks]

[Total for this question: 31 marks]

[Space for working continues...]

- $$X_t := \int_0^t s^\alpha dB_s.$$

[7 marks]

If you have used additional space for working then please tick here: ☐

[Total for this question: 11 marks]

4. Suppose $T, r, \sigma > 0$ are constants. Let $(W_t)_{t \geq 0}$ be a Brownian motion. Consider the following stochastic differential equation:

$$dS_t = rS_t dt + \sigma S_t dW_t, \quad (2)$$

and $S_0 > 0$ is a positive real number.

- (a) By applying Itô's formula to $Y_t = \ln S_t$, prove that the solution of (2) is

$$S_t = S_0 \exp \left(\left(r - \frac{1}{2} \sigma^2 \right) t + \sigma W_t \right). \quad (3)$$

[7 marks]

If you have used additional space for working then please tick here: ☐

(b) Using (3), show that for fixed $K > 0$,

$$\mathbb{P}(S_T > K) = N(d_-), \quad (4)$$

where $N(x)$ is the cumulative distribution function (CDF) for the standard normal distribution $N(0, 1)$ and

$$d_- = \frac{\ln\left(\frac{S_0}{K}\right) + \left(r - \frac{\sigma^2}{2}\right)T}{\sigma\sqrt{T}}.$$

[7 marks]

If you have used additional space for working then please tick here: ☐

(c) Using (3), show that for every $K > 0$,

$$\mathbb{E}[S_T 1_{\{S_T > K\}}] = S_0 e^{rT} N(d_+), \quad (5)$$

where $N(x)$ is the cumulative distribution function (CDF) for the standard normal distribution $N(0, 1)$ and

$$d_+ = d_- + \sigma\sqrt{T} = \frac{\ln\left(\frac{S_0}{K}\right) + \left(r + \frac{\sigma^2}{2}\right)T}{\sigma\sqrt{T}}.$$

[9 marks]

If you have used additional space for working then please tick here: ☐

- (d) Suppose that the stock price at time t is given by the stochastic process S_t in (3), and $S_0 > 0$ is the current price of the stock. The bank account pays interest continuously in time at a risk-free interest rate $r > 0$.

Using (4) and (5), compute the Black-Scholes price (or arbitrage-free price) $\Pi_0(\text{Call}(T, K))$ of the European Call option with strike K , maturity T , and payoff $\Phi(S_T) = (S_T - K)^+$. [5 marks]

[Total for this question: 28 marks]

If you have used additional space for working then please tick here: ☐

This is an extra page for working. Please indicate clearly the question number to which your working relates, otherwise your working may not be marked.

This is an extra page for working. Please indicate clearly the question number to which your working relates, otherwise your working may not be marked.

This is an extra page for working. Please indicate clearly the question number to which your working relates, otherwise your working may not be marked.

This is an extra page for working. Please indicate clearly the question number to which your working relates, otherwise your working may not be marked.